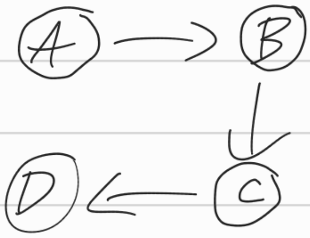
5. óra

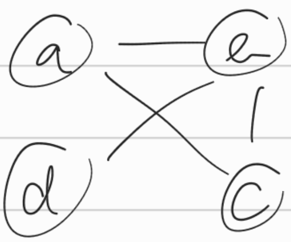


$(A,B) \in E$
$(B,C) \in E$
$(C,D) \in E$

$\langle A,B,C,D \rangle$ egy út

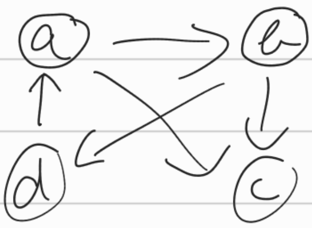
Gráfábrázolások:

– grafikus, szöveges:



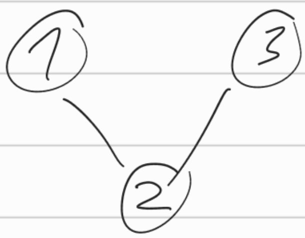
honnan — hova

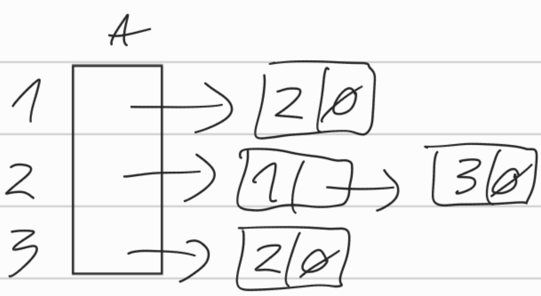
a – b; c
b – c; d



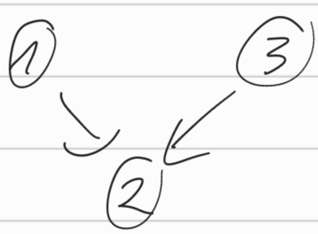
a → b; c
b → c; d
d → a

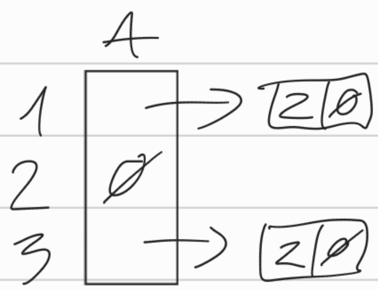
– éllista:



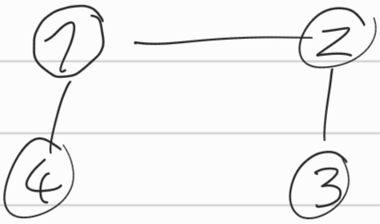


Edge
+ v : N
+ next : Edge*



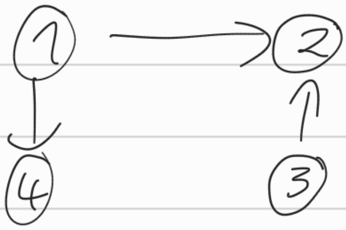


– csúcsmátrixos:



	1	2	3	4
1	0	1	0	1
2	1	0	1	0
3	0	1	0	0
4	1	0	0	0

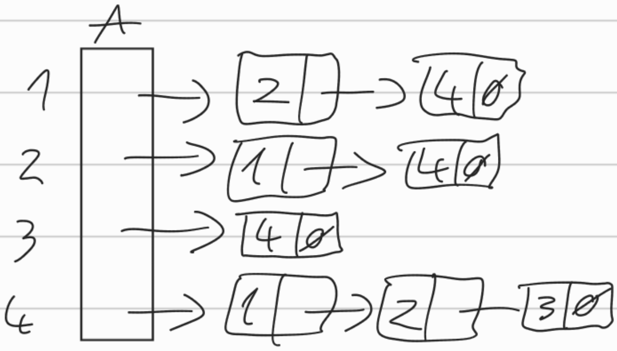
$A[i,j]$

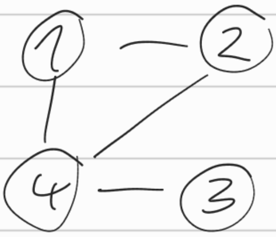


	1	2	3	4
1	0	1	0	1
2	0	0	0	0
3	0	1	0	0
4	0	0	0	0

Példa:

1 — 2;4
2 — 4
3 — 4

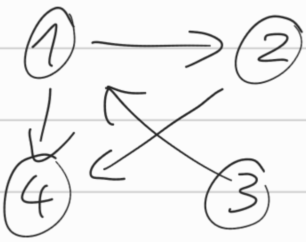


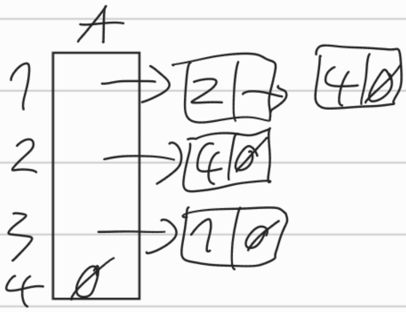


	1	2	3	4
1	0	1	0	1
2	1	0	0	1
3	0	0	0	1
4	1	1	1	0

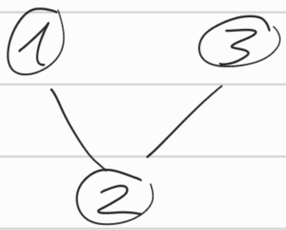
1 → 2;4
2 → 4
3 → 1

	1	2	3	4
1	0	1	0	1
2	0	0	0	1
3	1	0	0	0
4	0	0	0	0

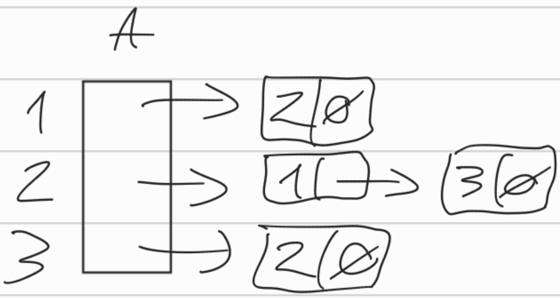


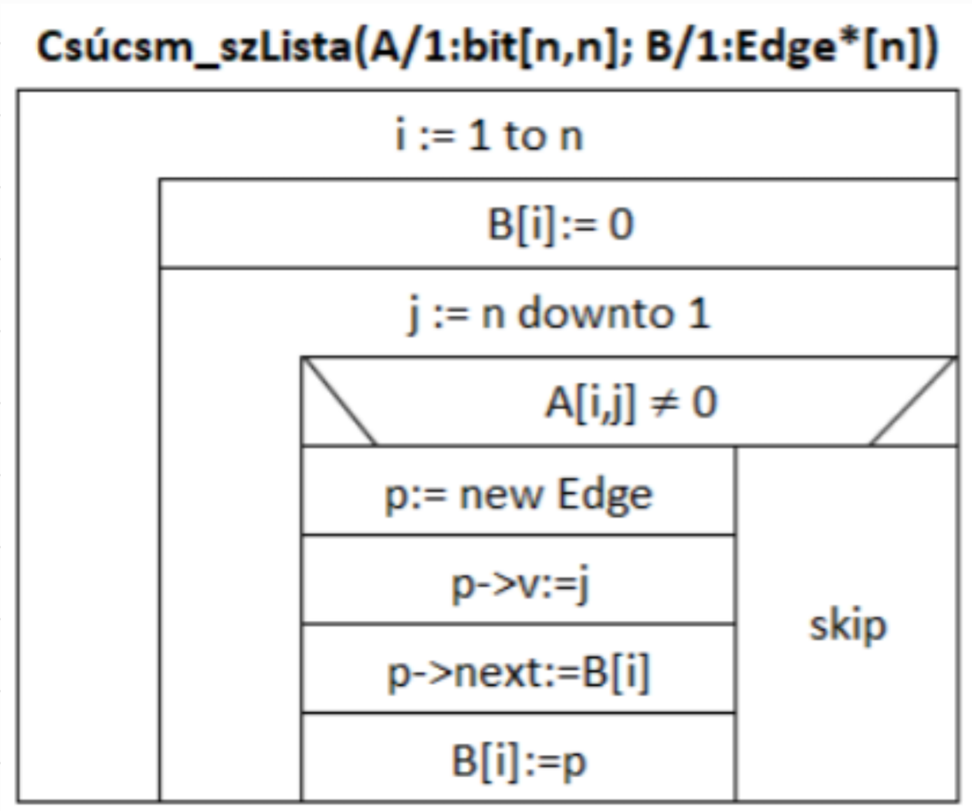


Csúcsmátrixból éllista:



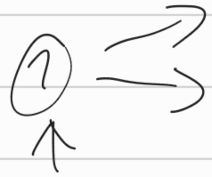
	1	2	3
1	0	1	0
2	1	0	1
3	0	1	0

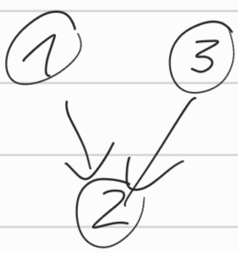


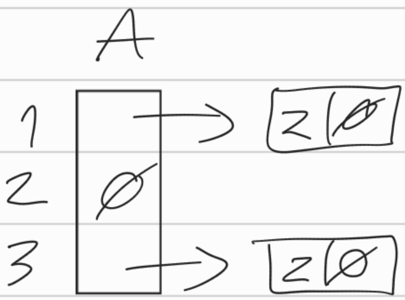


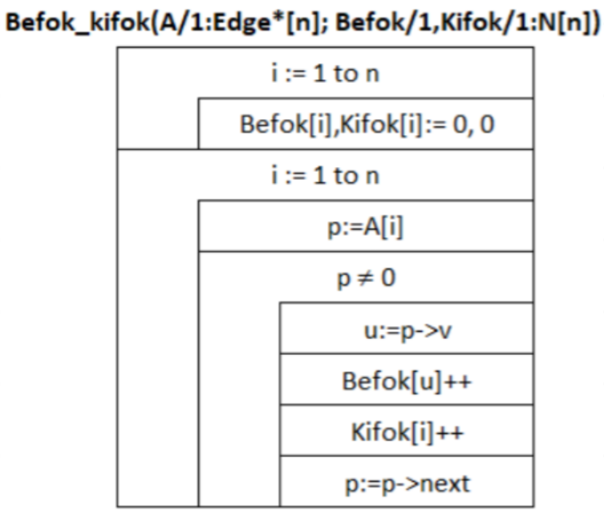
Befok - kifok:

befok: 1, kifok: 2

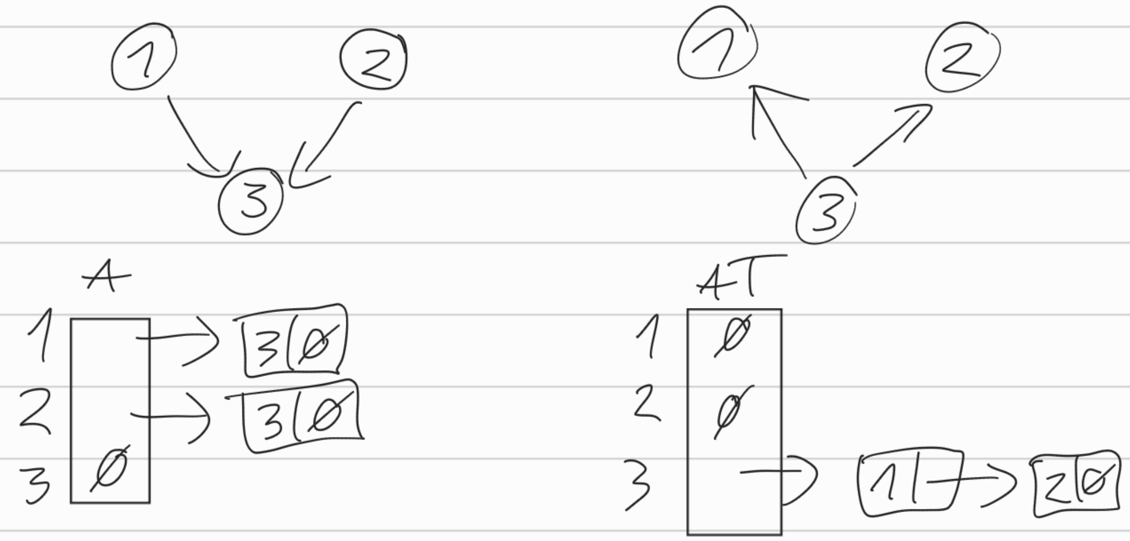








Transzponált gráf:



Transzponál(A/1,AT/1:Edge*[n])

```
i := 1 to n
    AT[i]:=0
i := n downto 1
    p:=A[i]
    p ≠ 0
        u:=p->v
        q:=new Edge
        q->v:=i
        q->next:=AT[u]
        AT[u]:=q
        p:=p->next
```

ZH:

1.) Huffman kódolás (14 pont)
2.) LZW kódolás és dekódolás (10 pont)
3.) AVL fa beszúrás és törlés (12 pont)
4.) B+ fa beszúrás és törlés (10 pont)
5.) Struktogram (14 pont)